IT6070 · ASSIGNMENT II · WEEK 1 PROGRESS

Reading the unreadable prescription

An AI-assisted prescription reader and dispensing check for a community pharmacist

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THE PROFESSIONAL

A pharmacist dispensing handwritten prescriptions

Registered Pharmacist
Dulnitha Pharmacy, Kottawa

Industry

Healthcare: independent community retail pharmacy in Kottawa, Sri Lanka

Daily role

Interprets and dispenses prescriptions, counsels patients, manages stock

Workload

About 200 prescriptions a day, 70–80% of them handwritten

Team and peak hours

3 staff; weekdays 7–8 am and 6–8 pm; busy all day at weekends

THE PROBLEM

Every handwritten prescription is decoded, cross-checked and matched to stock by hand: slowly, from memory, with a queue waiting.

1 · Decipher

Cursive handwriting and shorthand such as bd, tds, nocte or 1×2 must be read correctly.

2 · Cross-check

Duplicate therapy, interactions and dose sense are checked mentally, item by item.

3 · Match stock

Prescribed brands are mapped to what is on the shelf, or to an equivalent generic.

EVIDENCE OF NEED

What the pharmacist told us

Method

1. Discussion with the pharmacist at Dulnitha Pharmacy, Kottawa, on 30/09/2026
2. Figures are the pharmacist's own estimates

~200

prescriptions handled on a typical day

70–80%

are handwritten, about 140–160 a day

2–3 min

to read, check and dispense each one one

~3%

need a colleague or the prescriber to confirm

REQUIREMENTS

Two things the solution must do

R1

Read the prescription

From a phone photo of a handwritten prescription, extract each item's drug, strength, dose, frequency and duration into an editable list, expanding shorthand and flagging uncertain fields.

Accepted when: list appears within 30 s; every low-confidence field is highlighted; the pharmacist can correct any field.

R2

Check before dispensing

Match each confirmed item against the pharmacy's stock list and a small rule list of duplicates and interactions, producing a dispensing checklist with warnings.

Accepted when: out-of-stock items show in-stock generic alternatives; alternatives; flagged pairs raise a warning; no patient name is sent or or stored.

TECHNOLOGY CHOICE

Multimodal AI vs conventional OCR

Criterion	Multimodal LLM (chosen)	OCR + rule-based parsing
Handwriting, shorthand (R1)	Reads cursive; expands bd, tds, nocte in context	Built for print; weak on doctors' handwriting
Structured output (R1→R2)	Returns JSON fields directly for the stock check	Raw text; needs custom parsing rules
Uncertainty	Can flag doubtful fields for review	Character confidence only, not meaning
Privacy, connectivity	Needs internet; patient name must be redacted	Runs fully offline on the counter PC

Decision: a multimodal LLM drives R1; deterministic stock and rule checks handle R2, so safety logic stays predictable.

PLANNED PROOF OF CONCEPT

Two connected features in workflow

Input
Prescription photo

→

Feature 1 · AI
Extract items with confidence flags

→

Human in loop
Pharmacist confirms or edits

→

Feature 2 · Rules
Stock, generics and safety check

→

Output
Dispensing checklist

Fallback

API error, timeout or unreadable photo → retake prompt or manual entry form; Feature 2 still runs.

Planned stack

[Python + Streamlit] web app, [vision LLM API], stock and rule lists as CSV (simulated data).

RISK AND NEXT ACTION

What could block this

Blockers and risks

Patient data in real prescriptions

Use redacted or synthetic samples; never send patient identifiers to the API.

A misread drug is a safety risk

Decision support only: the pharmacist confirms every item before dispensing.